COLLEGE MANAGEMENT WEB

Full Project Guide & Test Plan

From first launch and account registration to admissions, academics, reports, and student record workflows.

Portal	Main responsibilities
Student	Register/sign in, submit admission application, upload documents, track notifications and academic results.
Admin	Review applications/documents, process admissions, manage records, view network/search/reports/audit.
Teacher	View department students, enter/publish marks, manage attendance/exams, review reports/profile.
Shared services	MySQL database, role sessions, file uploads, notifications, activity/audit records.

How to use this guide: Start with Section 1, then follow the student-to-admin workflow in order. Sections for teachers and data structures can be tested independently. Each test gives an action and an expected result.

Prepared for the current project files and local demo data. The project uses PHP, MySQL/MariaDB, and XAMPP.

1. Start the project and confirm the database

Local XAMPP setup

- Open XAMPP Control Panel and start **Apache** and **MySQL**.
- Open `http://localhost/CollegeManagementWeb/`. The root index redirects to the login page.
- The app connects using `config/database.php`. Defaults are database `college_management`, host `127.0.0.1`, user `root`, blank password, and candidate ports 3306/3307/3308. Environment variables DB_HOST, DB_USER, DB_PASSWORD, DB_NAME, and DB_PORT override defaults.
- On request, `config/schema.php` creates or upgrades the app tables. Confirm MySQL is running before opening the app.
- If Apache is on a non-default port, include it in the URL, for example `http://localhost:8080/CollegeManagementWeb/`.

Current database snapshot

Table/data	Rows at guide creation	Notes
students	14	Includes existing accounts and the 10 DS26 demo accounts.
student_applications	11	10 DS26 demo applications: 5 Accepted and 5 Pending; one existing application.
teachers	4	3 DS-TEACH demo teachers plus one pre-existing teacher.
subjects / exam_schedule	0 / 0	Teacher marks/exam pages need academic setup records before meaningful end-to-end tests.
student_documents	0	Upload documents during testing; admin verification is a separate step.
student_connections	7	Includes sample relationships; the lab filters graph to approved students.

The counts are a snapshot, not a promise that your database will always match. Your local records may change as you test.

Demo sign-ins

Role	Login identifier	Demo password
Student	Accepted: DS26-001 to DS26-005. Pending: DS26-006 to DS26-010. Use the roll number.	Use the local COLLEGE_DEMO_PASSWORD value set during seeding.
Teacher	sahana.mehta@example.test; anil.kumar@example.test; leena.joseph@example.test	Use the local COLLEGE_DEMO_PASSWORD value set during seeding.
Admin	Use the admin username and password already configured in your database.	Not printed here; use your private admin credential.

Demo-password handling: The seed script requires a local COLLEGE_DEMO_PASSWORD value. Do not store or share it. The login form authenticates teachers by email, not teacher code.

2. Login, registration, and role access

One entry page serves all roles

On `login.php`, choose Student, Admin, or Teacher sign-in. Student login uses a roll number; admin login uses a username; teacher login uses an email. Passwords are verified against stored hashes. A successful login creates a role session and redirects to that role's dashboard.

Student registration test

- On the login screen, open the student registration form.
- Use a fresh roll number in the required pattern `26mcaa45` (2 digits + 4 letters + 2 digits). Do not use a roll number already in the database.
- Enter a name of letters/spaces up to 20 characters; choose an allowed department (MSC, MCA, BCA, BSC, LAW, CSE), semester I-VIII, 10-digit phone, CGPA formatted like 8.93 in the 0-10 range, and matching password fields (minimum 6 characters).
- Submit. Expected: registration success message. Sign in with the new roll number and password; expected destination is Student Dashboard.

Negative registration tests

Try	Expected result
Reuse the same roll number	Duplicate registration is rejected.
Use an invalid roll format or phone not exactly 10 digits	Validation message; no new account.
Use mismatched confirmation or a password shorter than 6	Validation message; no new account.
Open a protected dashboard while logged out	Redirect to the corresponding login page.

Important distinction: Registration creates a student account in the students table; it does not itself mean an admission application has been accepted. The student must submit an application, then the admin reviews it. Demo DS26 accounts were seeded directly for testing and do not match the public registration form's roll-number pattern.

3. Student portal: application through progress

Student Dashboard is the student's starting page. It summarizes profile, application/admission readiness, documents, notifications, attendance/academic status, and links to the relevant pages. Admission Timeline shows the student's workflow status.

End-to-end student test

• Sign in with DS26-006 and the demo password. Open Student Dashboard and confirm the account sees its own Pending application state.

• Open Student Application. Complete the application fields (program type UG/PG, course, previous qualification/institution/percentage, address, city/state/pincode) and submit. Verify the application appears in the student's application history/timeline.

• Before admin accepts the application, open Student Documents. Expected: document upload is locked with an explanation; do not expect pending students to submit documents.

• Sign out and sign in as admin; accept the application (see Section 4). Sign back in as the student and confirm the updated status/notification.

• After acceptance and verification, upload required documents (PDF/JPG/JPEG/PNG; maximum 5 MB per file) under the correct document type. Confirm each file appears in the document list.

• Sign in as admin and verify/reject a submitted document. Sign back in as student and check notification/status.

• After all required uploaded documents are verified, admin can complete admission. Confirm timeline/dashboard reflects the admission number/status.

• When teachers publish marks for the student's subject, open Student Marks and confirm published academic results display.

Student pages and what to check

Page	Purpose / test
student_dashboard.php	Status cards, next action, document readiness, marks, attendance, notifications and profile photo.
student_application.php	Submit/update the admission application; course choices depend on program rules.
student_documents.php	Upload required documents after application acceptance/verification; test valid file and over-limit file.
admission_timeline.php	Read progress stages and current state.
notification_center.php	Open notices, mark one read, then mark all read; unread count should update.
student_profile.php	Review the student's own profile information.
student_marks.php	View marks after a teacher has published them; drafts should not be treated as published results.
student_network.php	Admin feature; not a student self-service graph.

Document upload is gated by an Accepted application with verification timestamp. The document verifier handles the admin decision; upload does not complete admission automatically.

4. Admin portal: review and manage

Sign in with the existing admin account. The admin dashboard is the navigation and workflow hub; sidebar entries open the modules below.

Module	How it works / test action
Applications - admin_applications.php	Filter/search student applications. Accept, reject, or reset an application. Accepting creates an admission-processing row and notifies the student. Test with a Pending demo application.
Document Verification - document_verification.php	Review documents for accepted applicants. Verify or reject with optional remarks. Confirm student notification and status.
Admission Processing - admission_processing.php	Move an eligible accepted application to Processing, Completed, or Rejected. Completion is blocked until at least one document exists and every submitted document is Verified.
Students - students.php / edit_student.php	List, add, edit, and delete student records. Test add/edit with a unique valid roll number. Take care: delete is destructive.
Teachers - teachers.php	Create teacher accounts, see roster, and toggle ACTIVE/INACTIVE. Login with an inactive teacher should be denied.
Departments - department.php	Review department information and related academic data available in this project.
Search & Sort - search_sort.php	Search by roll/name and sort by name or CGPA. This project version displays accepted-application students only.
Student Network - student_network.php	Create/remove valid relationships; students must exist, differ from each other, and share department or semester. Graph edges are persisted.
Reports - reports.php	Inspect available reports and filters; compare displayed totals to the records in the current database.
Activity Log - activity_log.php	Review logged application and administrative activity after completing test actions.
Admin Photo Management - admin_photo_management.php	Find a student/teacher and upload/replace/delete a profile image. Allowed image formats JPG/JPEG/PNG/WEBP, max 5 MB.
Student Record Workflows - data_structures_showcase.php	Database-fed student review and academic-planning tools; details and tests in Section 8.

Application review test

- Open Applications and locate DS26-006 (Pending). Use Accept. Expected: status Accepted, a student notification, and an admission-processing record.
- Repeat with DS26-007 using Reject, then confirm Rejected state. Use reset-to-pending only if you need to repeat this test.
- Return to student login after acceptance and verify its application state changed.
- Test duplicate/invalid selection controls without deleting any real records.

Avoid delete tests on production/important records. Use only the DS26 demo cohort for destructive actions and make a database backup first.

5. Documents and admission completion rules

Admission is a sequence of separate states. Application acceptance is not the same as completed admission.

Stage	Owner	Expected transition
Application submitted	Student	Application is recorded; admin sees it for review.
Accepted / Rejected	Admin	Accepted opens the document workflow; rejection stores the decision and notification.
Documents uploaded	Student	Required document types are associated with the student's application.
Documents Verified / Rejected	Admin	Each file is reviewed and its status/remarks are saved.
Admission Processing	Admin	An accepted application can move to Processing.
Completed	Admin	Allowed only when at least one document is present and all documents for the application are verified; an admission number is generated.

Test a successful completion

- Use an accepted applicant such as DS26-001. Upload at least one allowed document with size below 5 MB.
- In Document Verification, locate the uploaded file and mark it Verified.
- In Admission Processing, choose Processing first if desired, then Completed. Expected: generated admission number and Completed status.
- Try to complete another accepted application without documents. Expected: completion is blocked with a clear requirement message.
- Try a file above 5 MB or an unsupported type. Expected: upload is rejected and no usable document row is created.
- Check the student's notification center and admission timeline after status changes.

The 5 accepted DS26 samples are application-approved but currently have no sample documents. The 5 pending DS26 applicants are not eligible for document upload or admission processing until admin acceptance. Testing upload/verification is a manual action in the portal.

Document security basics to verify

- Use safe test documents, never upload confidential identity documents into a shared demo system.
- Uploaded files are placed under uploads/student_documents; confirm the file appears only against the correct student/application.
- Do not use arbitrary executable files as test payloads. A harmless unsupported text file is enough to check rejection.

6. Teacher portal: classes and academic work

Teacher accounts sign in by email. Only ACTIVE accounts can authenticate. Teacher pages are scoped to the signed-in teacher and department; use the sample emails from Section 1.

Page	Testing path
teacher_dashboard.php	Check teacher identity and summary cards, then use the action tiles.
teacher_students.php	View department/assigned student cards and open a student's academic context.
teacher_marks.php	Select a department student, semester, and subject. Enter valid midterm/end-semester marks. Save as draft, then publish; verify published marks in the student portal.
teacher_attendance.php	Select date/session, set each student's status (present/late/absent), save, refresh, and confirm values persist.
teacher_examinations.php	Review scheduled exams relevant to the teacher's department/subjects.
teacher_reports.php	Review available academic performance reports and filters.
teacher_profile.php	View/update the current teacher profile as supported by the form.

Prerequisite in the current database

At guide creation, there are no rows in subjects or exam_schedule. Consequently, Marks Entry and Examination pages may show empty selectors or no exams until academic setup data is present. The sample cohort seed creates teachers and student/application data, not subjects, exams, marks, or attendance. Do not interpret an empty academic list as a login failure.

Teacher test when academic rows exist

- Use an active teacher account whose department matches a configured subject and approved/admitted students.
- Enter marks within the displayed allowed maximum. The page uses a midterm/end-semester split and saves a draft or publishes.
- Test out-of-range marks; expected: validation error and unchanged stored marks.
- Mark attendance for an available scheduled session, save, reload, and check persistence.
- Log in as the student; drafts should not be visible as published results, while published results should appear.
- Set a teacher account inactive from Teachers, log out, then test login. Expected: account-deactivated message.

Do not fabricate academic records just to make the screens look full. Configure actual test subjects/schedules through your institution's chosen setup process, then use the portal's normal flows.

7. Dashboard, reports, photos, and notifications

Admin dashboard

Use dashboard.php to check counts and the Pending Action Center. After accepting an application, uploading a document, or completing admission, refresh and confirm the relevant cards/counts move consistently. The dashboard summarizes records; it is not the source of the workflow action.

Reports and activity

- Open Reports and try each visible report tab/filter. Search a known demo roll number where supported and compare report data with the student record.
- After testing application acceptance/rejection, inspect Activity Log for corresponding admin actions and timestamps.
- Open Notification Center as a student, mark one notice read, then mark all read. Confirm unread indicator/count changes.

Profile photo test

- Admin opens admin_photo_management.php and searches a demo student or teacher.
- Upload a non-sensitive JPG, PNG, or WEBP image under 5 MB. Verify preview and success state.
- Log in as that user and verify photo appears on the dashboard. Replace it, then delete it and confirm the placeholder returns.
- Try a file larger than 5 MB and an unsupported type; confirm rejection. The feature is admin-only.

Sample cohort sign-ins for end-to-end testing

Accepted accounts	Pending accounts
DS26-001 Aarav Nair - CSE / Sem 1 / CGPA 8.71 DS26-002 Diya Sharma - CSE / Sem 2 / CGPA 9.14 DS26-003 Ishan Rao - CSE / Sem 3 / CGPA 7.82 DS26-004 Mira Das - CSE / Sem 4 / CGPA 8.46 DS26-005 Neel Menon - CSE / Sem 1 / CGPA 9.02	DS26-006 Riya Shah - CSE / Sem 1 / CGPA 7.54 DS26-007 Kabir Sen - CSE / Sem 2 / CGPA 8.11 DS26-008 Tara Iyer - CSE / Sem 3 / CGPA 7.96 DS26-009 Dev Patel - CSE / Sem 4 / CGPA 8.32 DS26-010 Anya Roy - CSE / Sem 1 / CGPA 8.04

Demo teacher	Login email	Department / designation
Sahana Mehta	sahana.mehta@example.test	CSE / Assistant Professor
Anil Kumar	anil.kumar@example.test	IT / Associate Professor
Leena Joseph	leena.joseph@example.test	MCA / Assistant Professor

The seeded student/teacher accounts use the local COLLEGE_DEMO_PASSWORD value set when running seed_course_demo.php. The guide does not contain that value. Teacher authentication uses email.

8. Student record workflows and academic planning

Open Student Record Workflows from the admin sidebar. This page queries database records: accepted students, pending application tokens, active teachers, and saved student_connections edges. Student names/details for Pending applications are intentionally not passed to the page.

Feature	How to test / expected result
Linked list	Use approved-student selector to insert, unlink, and traverse. Unlink changes the in-page list only; it never deletes the DB student.
Stack	Push an approved student several times, then Pop. Last pushed item is removed first.
Queue	Enqueue pending application-number tokens, then Dequeue next. Front token leaves first. Status in the admissions table is not changed.
Expression stacks	Evaluate (8 + 4) * 3 / 2. Expected result 18; test 1 / 0 for a guarded error.
Binary tree	Tree is built from accepted students. Inspect inorder, preorder, postorder roll sequences; no database write.
BST	Search an exact accepted roll number such as DS26-003. The page displays found/missing and its comparison path.
Phase 3: Find and organize approved students for admission decisions (CLO2)	Accepted student snapshot only: partial lookup checks each name/roll in turn; exact lookup orders a copy and repeatedly checks the middle roll number, narrowing to one half. Sorting compares roll/name as text or CGPA numerically; insertion ordering shifts into an ordered prefix, while merge ordering splits and merges sorted copies. Results and comparison counts are displayed without database writes.
Graph/BFS	Adjacency list and matrix use saved student_connections restricted to accepted vertices. Choose BFS start; only connected component is visited.
Phase 4: Plan teacher support for accepted students (CLO3)	The page combines Accepted student records with active teachers. Advising assigns one student at a time to the least-loaded same-department teacher when possible. Tutoring fills a student-by-budget score table and traces it back to choose the highest-need set within 1-50 hours. Adviser planning recursively tries teachers, enforcing ceiling caseload and unique semesters per teacher; conflicts undo a tentative assignment and try the next teacher. Proposals are read-only and not saved.

These algorithms run against the data loaded for the page and are not a replacement for persistent records. The database-backed relationship editor remains Student Network. Search & Sort links to the existing project search workflow.

9. Complete end-to-end test run

Use demo records only. Keep the accepted/pending test distinction intact; do not complete or reject all sample applications if you still need them for demonstrations.

#	Action	Expected checkpoint
1	Start MySQL and Apache; open project URL.	Login page appears without DB connection error.
2	Sign in as a demo pending student DS26-006.	Student dashboard opens; own Pending application is visible.
3	Open documents while application is Pending.	Upload is locked.
4	Sign out; sign in as admin; accept DS26-006.	Application Accepted, notification created, processing row available.
5	Sign in as DS26-006; upload allowed file under 5 MB.	Document is listed as pending verification.
6	Admin verifies the document.	Document status changes and student is notified.
7	Admin completes admission.	Admission number is generated; completion is blocked unless all uploaded docs are verified.
8	Open admin Search & Sort, Student Network, and Student Record Workflows.	Student names appear only after an Accepted application; graph/search use accepted cohort.
9	Use a demo teacher sign-in.	Teacher dashboard opens. Academic functions need configured subjects/exams.
10	Test invalid login, duplicate registration, invalid file, invalid marks if academic data exists.	Validation feedback appears; invalid operation does not persist.
11	Sign out and revisit protected page URL.	Page redirects to login.

Resetting the demo state

There is no one-click reset script because deleting or resetting admission, document, attendance, or academic records can destroy useful work. Before the demo, record the application IDs/statuses and use admin actions to restore only the specific DS26 test records you changed. Never run broad DELETE statements against the database.

10. Troubleshooting, safety, and project map

Symptom	Check
Database unavailable	Start MySQL; check DB_HOST/DB_PORT/DB_NAME and credentials in config/database.php; try phpMyAdmin connection.
Login rejected	Use role-specific identifier: student roll number, admin username, or teacher email. Demo password is case-sensitive.
Pending person not in DSA/search/network	Expected: student-based features require an Accepted application. Review status in Applications.
Documents are locked	Application must be Accepted and verified; student needs an active application.
Cannot complete admission	At least one document must exist and all documents for that application must be Verified.
Teacher marks/exam selector empty	Current database has no subjects and no exam schedules. Configure academic setup data first.
Teacher cannot see student	Check matching department and admission eligibility/status for that feature.
Photo/document rejected	Check supported extension, actual file type, and 5 MB limit; check upload folder permissions.

Safety before testing

- Back up the database before testing deletes, status changes, or file removals.
- Use the DS26 cohort and harmless files, not real student personal documents.
- Do not share the demo password publicly; replace it for any non-local use.
- Only test actions using role accounts you are authorized to use.

Where the main implementation lives

Area	Files
Entry/auth	index.php, login.php, logout.php, logout_teacher.php, config/database.php, config/schema.php
Student journey	student_dashboard.php, student_application.php, student_documents.php, admission_timeline.php, notification_center.php, student_profile.php, student_marks.php
Admin workflow	dashboard.php, admin_applications.php, document_verification.php, admission_processing.php, students.php, teachers.php, department.php
Search/network/insight	search_sort.php, student_network.php, reports.php, activity_log.php, data_structures_showcase.php
Teacher portal	teacher_dashboard.php, teacher_students.php, teacher_marks.php, teacher_attendance.php, teacher_examinations.php, teacher_reports.php, teacher_profile.php
Photos / local demo seed	admin_photo_management.php, photo_handler.php, config/photo_utils.php, seed_course_demo.php

This guide explains the visible flows and the state checks that matter for testing. Use the source files and the live database as the final authority if UI wording or counts change.